A3-PS002-TC028

Personal Consent Management Browser Extension

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AIORI Hackathon 3

October 13, 2026

Problem Description

Problem addressed and key issues

- Privacy notices are often long and difficult to understand, so users may agree without knowing what data is collected or why.
- Consent is scattered across websites, making it hard to review or manage choices in one place.
- Users may not have a clear way to withdraw consent after granting it.
- A recorded withdrawal does not guarantee that supported downstream actions stop.
- Changes to privacy policies can affect existing choices without clear, timely explanations.

We propose a browser extension that simplifies notices, records purpose-specific decisions, and monitors supported actions against the user's current preferences.

Background

Background and motivation

Websites commonly use privacy notices, cookie banners, and permission prompts to request user agreement. These interfaces may bundle several purposes together and present information in lengthy legal language.

The problem statement focuses on purpose-based consent evidence and enforcement: a service should record the purpose, notice version, grant or withdrawal, and check the current state before allowing a protected request.

Our browser-extension concept applies this idea to supported websites through a central consent dashboard. It will provide simplified summaries, granular choices, policy-change alerts, and a traceable history. Enforcement will be limited to actions the prototype can reliably observe and control.

Literature & RFC Review

Relevant standards and design considerations

Data-protection frameworks emphasise informed choice, purpose limitation, and the ability to withdraw consent. The EU General Data Protection Regulation (GDPR), Article 7, addresses conditions for consent and withdrawal. India's Digital Personal Data Protection Act, 2023, establishes a framework for processing digital personal data.

Browser extensions can inspect supported page content, but access depends on browser permissions and website behavior. The prototype should request only necessary permissions and distinguish observed events from actions it cannot control.

Our design combines purpose-specific records, notice-version hashes, a central dashboard, and a mock enforcement layer. It demonstrates technical evidence, not universal legal compliance.

Proposed Solution

Proposed solution: Personal Consent Management Extension

1. **Consent detection:** identify supported cookie banners, permission prompts, and privacy notices.
2. **Three-point summary:** explain **what data is collected, why it is collected, and how it will be used.**
3. **Granular controls:** let users accept or reject individual purposes rather than bundled choices.
4. **Consent ledger:** record grant, rejection, withdrawal, timestamp, purpose, and notice-version hash.
5. **Central withdrawal:** let users review and revoke supported consent decisions from one dashboard.
6. **Enforcement and alerts:** block or flag supported conflicting actions, record the outcome, and show the reason.
7. **Policy monitoring:** detect changes on supported policy pages and notify users when re-review may be needed.
8. **Email preferences:** optionally filter promotional messages through a supported email-provider integration.

The prototype will use synthetic accounts and test websites. Every enforcement result will be logged; unsupported actions will be labelled as not enforceable rather than falsely reported as blocked.

Optimization Proposed

Optimization proposed: before and after

Area	Before	With proposed extension
Understanding	Long notices	Three-point summary
Choice	Bundled acceptance	Purpose-specific controls
Record keeping	Scattered across sites	Central consent history
Withdrawal	Hard to locate	Central dashboard
Enforcement	May be unclear	Supported actions checked
Policy updates	Manual rechecking	Change alerts and review
Evidence	Limited visibility	Timestamped audit trail

Scope note: enforcement and email filtering depend on browser permissions, supported sites, and available integrations.

Solution Architecture & Design

Browser extension, consent registry, enforcement gateway, and audit services.

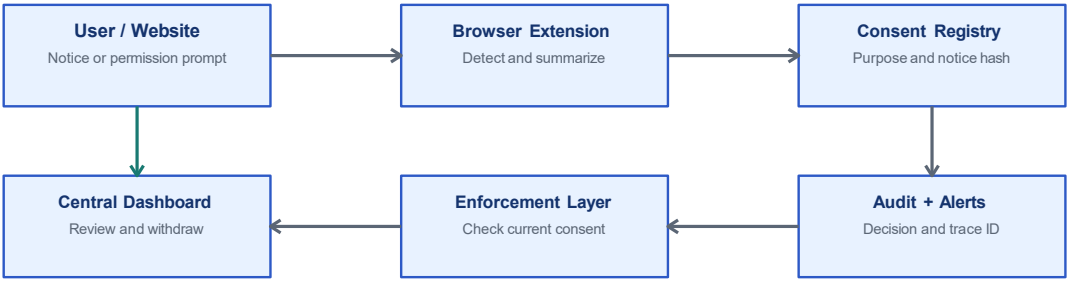


Figure: ConsentGuard System Architecture

Data Flow Diagram

From notice detection to purpose-specific decisions, enforcement, and audit logging.

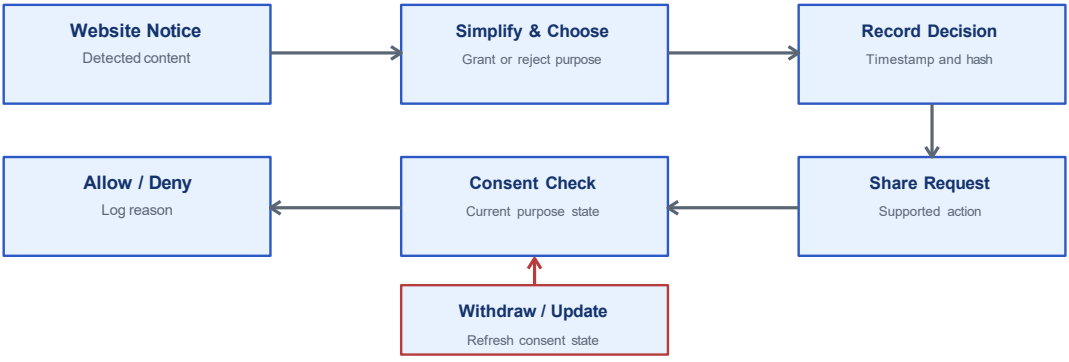


Figure: Consent Decision Data Flow

Deployment Timeline

Days 1-2: Set up the repository, React interface, FastAPI backend, and SQLite schema.

Days 3-4: Implement purpose-specific grants, rejections, withdrawals, notice versions, and hashes.

Days 5-6: Build the mock sharing API and enforce current consent on every protected request.

Days 7-8: Connect the dashboard, access monitor, and audit history.

Days 9-10: Add policy-change checks and downstream action tracking.

Days 11-12: Add email filtering only if provider integration is available; complete at least 12 lifecycle tests.

Days 13-14: Run end-to-end demonstrations, fix issues, export audit evidence, and prepare documentation.

Bibliography

[1] European Union. Regulation (EU) 2016/679 (General Data Protection Regulation), Article 7: Conditions for consent. 2016.

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[3] Google. Chrome Extensions Documentation: Declare permissions and host permissions. *https://developer.chrome.com/docs/extensions/develop/concepts/declare-permissions/*

[4] World Wide Web Consortium (W3C). Tracking Preference Expression (DNT). W3C Recommendation, 2019.

Thank You for Your Attention!